ADSDB P1 / UPC

EcoSentinel

Wildlife evidence, connected

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Photo: lynxexsitu.es / CC BY 3.0 es

A photograph starts the investigation

“What can help me understand this photograph?”

Related images
Recordings
Source descriptions



Education and biodiversity exploration

Photo: Alexkom000 / CC BY 4.0

Three modalities, one reference corpus

198

accepted assets
10 Iberian wildlife species

100 Images / Wikimedia

34 Audio / Xeno-canto

64 Text / Wikipedia + GBIF

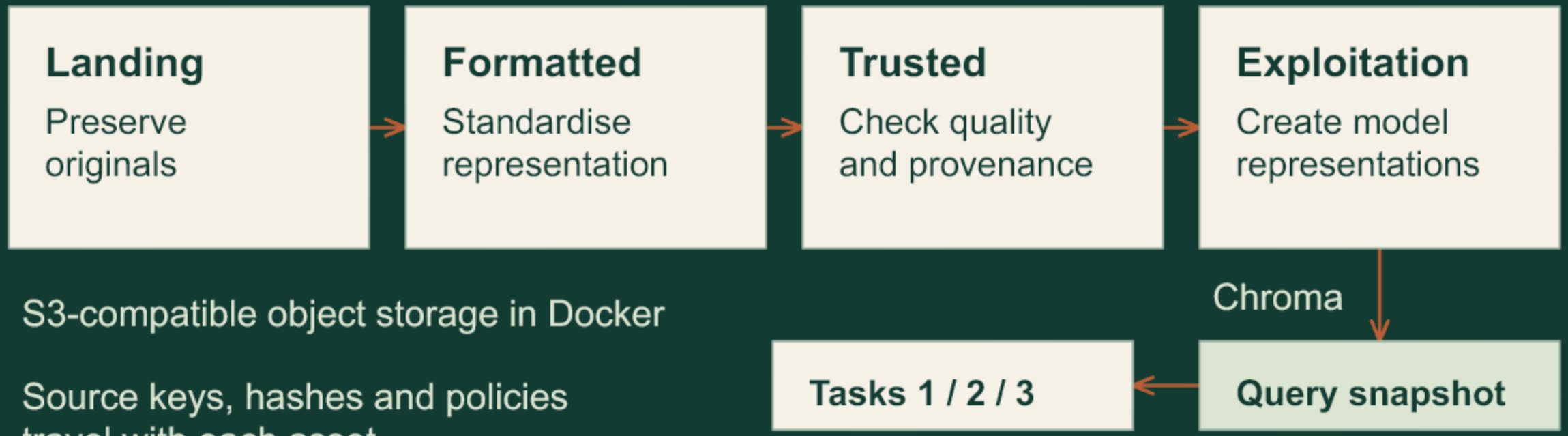
Uneven availability remains visible: no audio for the ocellated lizard

The EcoSentinel pipeline

Optional acquisition or local files / Security Gate

16-step orchestrator

Lock · timeouts · per-step reports

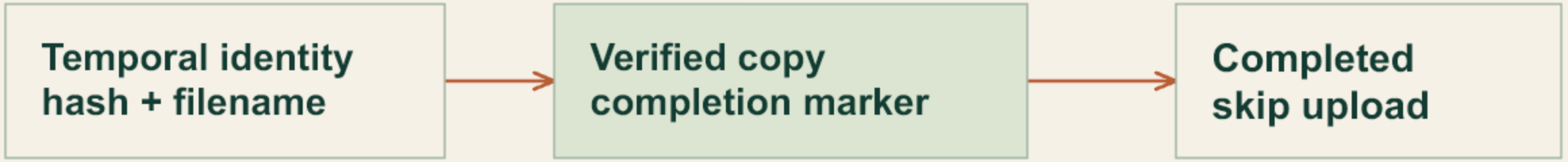


S3-compatible object storage in Docker

Source keys, hashes and policies travel with each asset

Landing recognises a completed ingest

Same bytes + same local filename



Changed bytes or name: a new ingest identity

Recorded rerun: 0 uploads / 3 completed assets

Formatted then standardises JPEG RGB / FLAC PCM16 / UTF-8 text

Formatted: representation by modality

Images

RGB JPEG / q = 95

Keep width + height
No crop or resize
White under transparency

Audio

FLAC / PCM16

Keep rate + channels
Keep full duration
No volume normalisation

Text

UTF-8 / LF / NFC

Keep language
Keep case + punctuation
Standardise Unicode

Originals stay in Landing / model-specific adaptation happens in Exploitation

Trusted: rejection or warning?

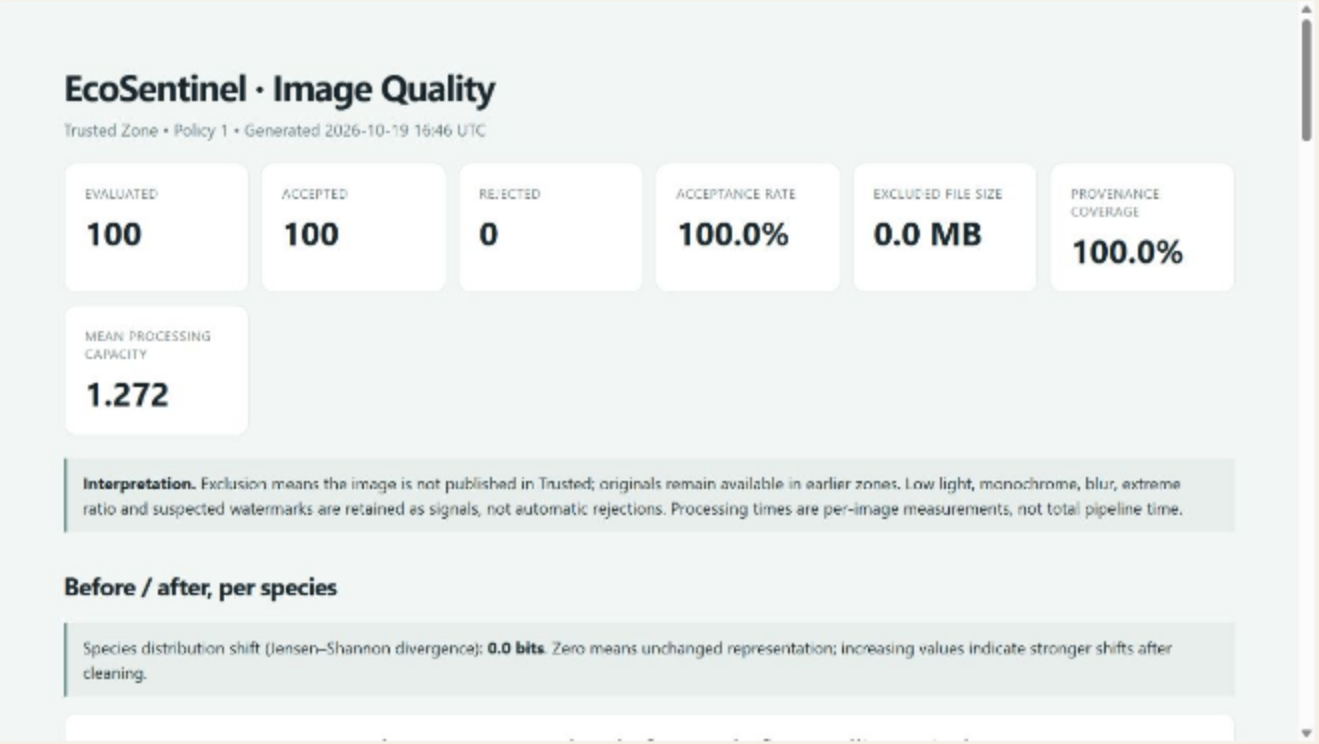
Observation	Policy
Invalid decoding	Reject with a reason
Confirmed image duplicate	Exclude the later matching image
Watermark / low-energy audio	Retain with a warning

Image budget: 20 MB / 40 MP / minimum 128 × 128

Audio budget: 200 MB / text: 4 MB / warnings preserve useful signals

Quality decisions leave an audit trail

Per-asset manifests / HTML + CSV + JSON + plots



EcoSentinel · Image Quality

Trusted Zone • Policy 1 • Generated 2026-10-19 16:46 UTC

EVALUATED	ACCEPTED	REJECTED	ACCEPTANCE RATE	EXCLUDED FILE SIZE	PROVENANCE COVERAGE
100	100	0	100.0%	0.0 MB	100.0%

MEAN PROCESSING CAPACITY
1.272

Interpretation. Exclusion means the image is not published in Trusted; originals remain available in earlier zones. Low light, monochrome, blur, extreme ratio and suspected watermarks are retained as signals, not automatic rejections. Processing times are per-image measurements, not total pipeline time.

Before / after, per species

Species distribution shift (Jensen–Shannon divergence): **0.0 bits**. Zero means unchanged representation; increasing values indicate stronger shifts after cleaning.

100%

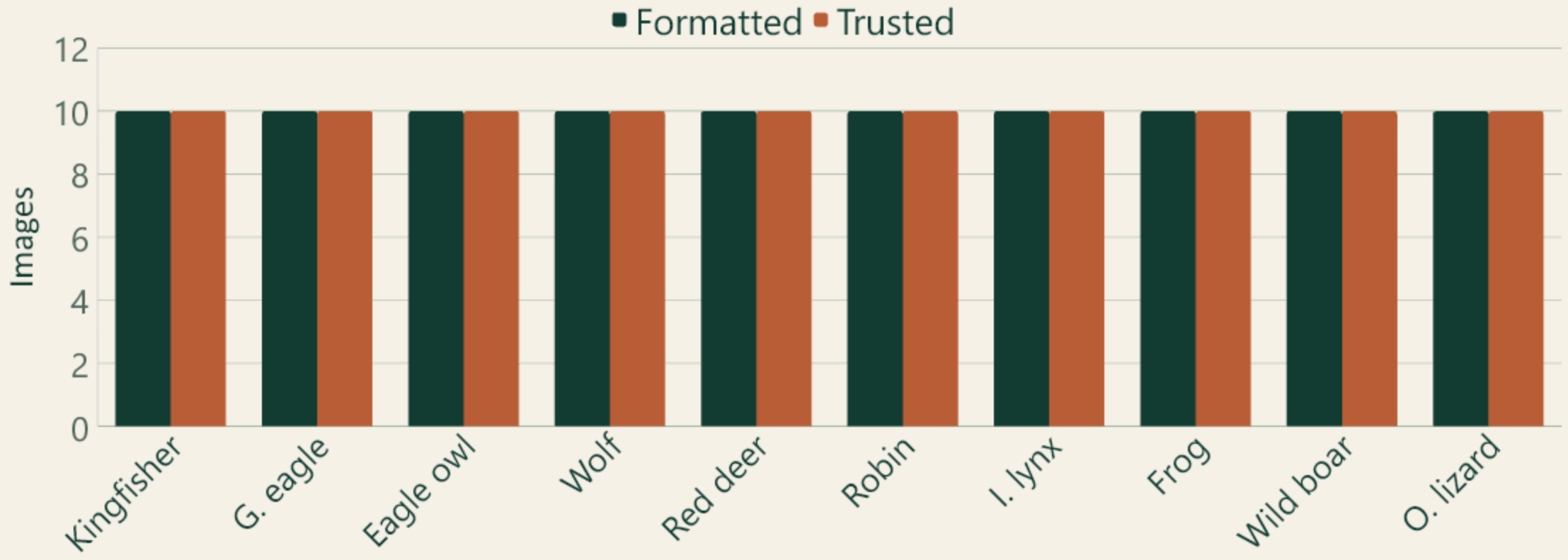
Provenance fields complete

0.0 bits

Species distribution shift

Real image report / 100 retained / 32 possible watermark flags / 19 Oct 2026

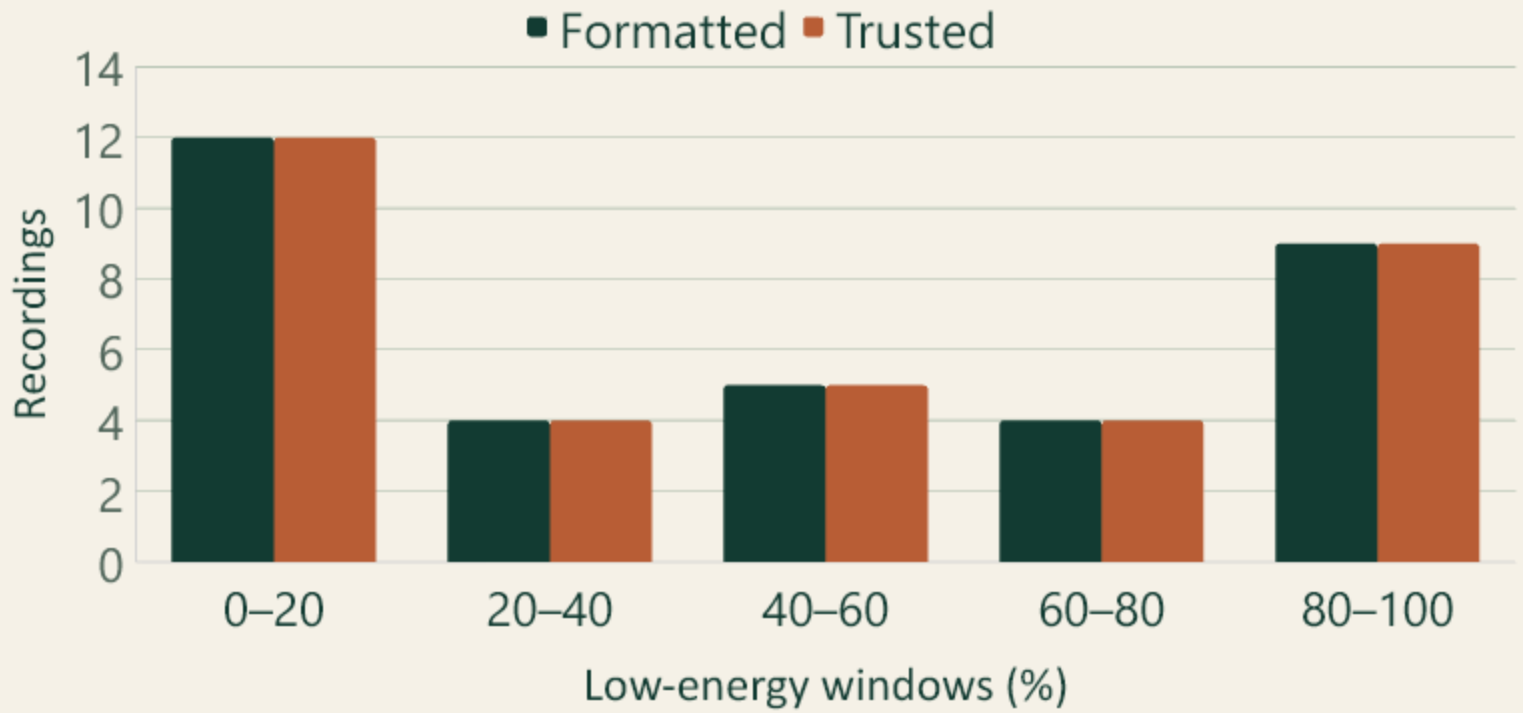
Did filtering change species representation?



100 images before / 100 after / Jensen–Shannon divergence: 0.0 bits

Quiet intervals remain in the evidence

Recordings grouped by low-energy window percentage



9 warnings

34 retained

Complete recordings remain available

50-ms windows below −50 dBFS / flag at ≥80% / same observed distributions

Five collections, 6,021 embeddings

6,021

embeddings
from 198 assets

Image / CLIP	**100**
Audio / CLAP	**357**
Text / E5	**547**
Image + text / CLIP	**1,763**
Audio + text / CLAP	**3,254**

Long recordings and documents produce multiple segments with explicit identities

Three encoders, three relationships



CPU execution / pretrained models / no comparative wildlife benchmark

Task 1: a real image query

REFERENCE IMAGE



Query / CLIP

SELECTED MATCHES / RANKS 1 AND 3



#1 / cosine 0.912



#3 / cosine 0.871

5 distinct assets / reference excluded / 4 lynx labels and 1 wolf label

Task 2: one query, images and text

“Iberian lynx in its natural habitat”

IMAGE HIT #5 / WIKIMEDIA



TEXT HIT #4 / GBIF

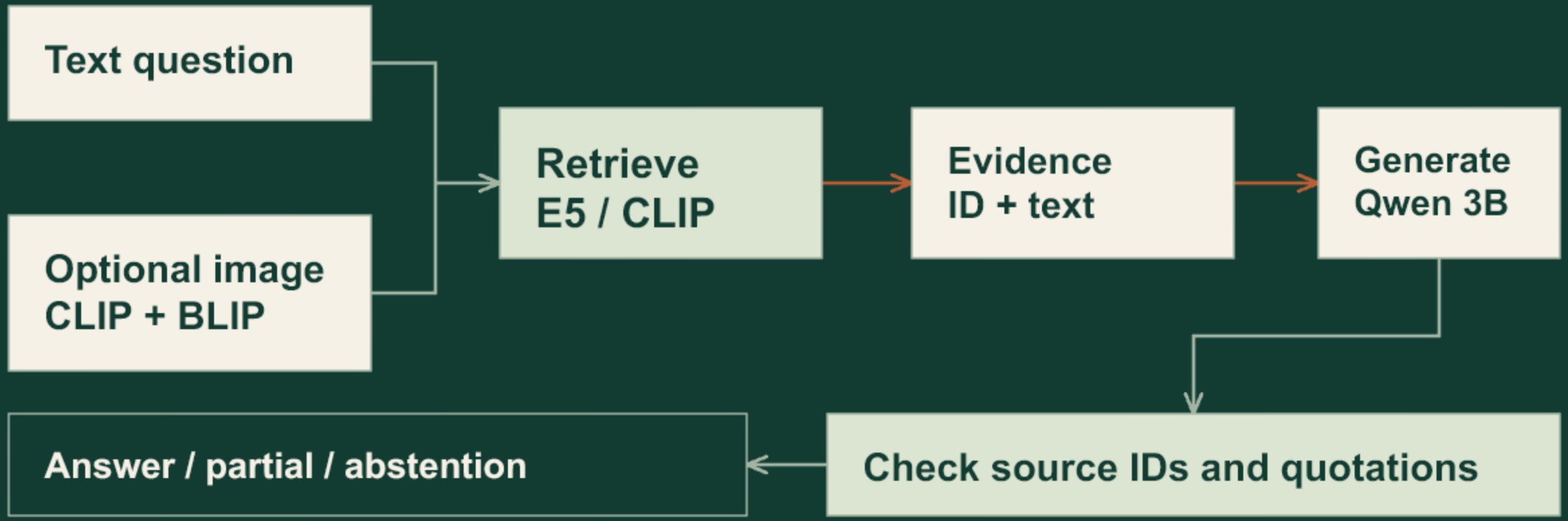
“About half the size of Eurasian Lynx”

Catalogue label: Lynx pardinus

8 results / 4 images + 4 texts / explicit diversity policy

Recorded CLAP example: “owl calling at night” / audio + text

Task 3: answer from retrieved evidence



Technical errors stay separate

Qwen receives text. Source matches do not prove semantic correctness.

The frog case: evidence and uncertainty

Corpus candidate

Pelophylax perezi

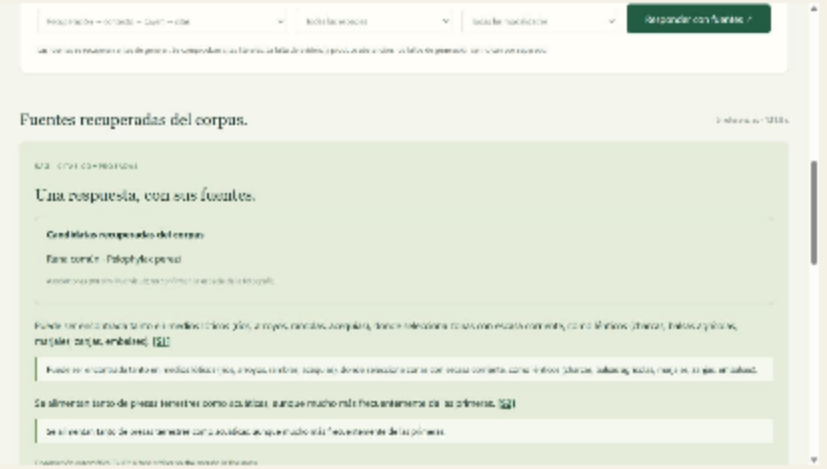
Habitat [S1]

“areas with little current”

Diet [S2]

“terrestrial and aquatic prey”

Actual interface / Spanish



Separate test: complete-genome request / abstained

Real model runs / translated extracts / species identity remains unconfirmed

The second run adds zero new vectors



16 steps / isolated storage / real CPU models

Reconciliation checks outcomes. This synthetic run is not a relevance benchmark.

An input can fail before it reaches storage

lynx.jpg

Harmless MZ header / synthetic fixture

BLOCKED

Before hashing or upload

Corrupt files / empty inputs
Archive expansion / resource limits

Watermark warning retains the image

364 offline tests + 1 integration / 71.82% coverage / 6,021 original vectors preserved

A query with a traceable answer

A result we can trace back to its source



Next: labelled retrieval and answer evaluation with specialist review